Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 7 Handout

Thursday, October 1

Problem 7.1

We revisit the classic medical test problem where a treatment decision must be made based on the results of a diagnostic test. Let random variable $X \in \mathcal{X} \triangleq \{\mathtt{h}, \mathtt{i}\}$ represent whether the patient is healthy ($X = \mathtt{h}$) or infected ($X = \mathtt{i}$). Suppose 1 patient in 100 is infected, i.e., $p_X(\mathtt{i}) = 0.01$. Meanwhile, we have a test that is 90% accurate. We use $Y \in \mathcal{Y} \triangleq \{\mathtt{-}, \mathtt{+}\}$ to represent whether the test result is negative ($Y = \mathtt{-}$) or positive ($Y = \mathtt{+}$), and $p_{Y|X}(\mathtt{-}|\mathtt{h}) = p_{Y|X}(\mathtt{+}|\mathtt{i}) = 0.9$.

(a) Suppose the test result is positive. Determine the probability that the patient is infected.

(b) The following table defines the cost $c(x, \hat{x})$ (in thousands of dollars) of deciding that the patient's state is $\hat{x}$ when it is x.

x	$c(x, \mathtt{h})$	$c(x, \mathtt{i})$
$\mathtt{h}$	0	1
$\mathtt{i}$	20	5

Determine the decision $\hat{x}$ that minimizes the expected cost if the test result is positive.

(c) Now suppose $c(x, \hat{x}) = \mathbb{1}(x \neq \hat{x})$. How does that affect you decision?

Now suppose we run the test n times, producing test results $Y_1, \ldots, Y_n \in \mathcal{Y}$. Assume that $Y_1, \ldots, Y_n$ are independent and identically distributed given the patient's state X, i.e.,

$$p_{Y_1,\ldots,Y_n|X}(y_1, \ldots, y_n|x) \triangleq \prod_{i=1}^{n} p_{Y|X}(y_i|x), \quad \text{for all } x \in \mathcal{X},\ y_1, \ldots, y_n \in \mathcal{Y}.$$

We define $k \triangleq \sum_{i=1}^{n} \mathbb{1}(y_i = \mathtt{+})$ to be the number of positive test results.

(d) Determine the maximum likelihood (ML) decision $\hat{x}_{\mathrm{ML}}(y_1, \ldots, y_n)$ given the observations $Y_1 = y_1, \ldots, Y_n = y_n$, expressing your answer in terms of k and n.

(e) Determine the maximum *a posteriori* (MAP) decision $\hat{x}_{\mathrm{MAP}}(y_1, \ldots, y_n)$ given the observations $Y_1 = y_1, \ldots, Y_n = y_n$, expressing your answer in terms of k and n.

(f) Determine the decision rule that minimizes the expected cost defined in part (b) based on the observations $Y_1 = y_1, \ldots, Y_n = y_n$, expressing your answer in terms of k and n.